Week 7 BA

2024 Aug04 Quiz-2

Comprehension for questions 1-8 (source Q2)

Milo's Evaluation (ME) is a loan default prediction company that uses manual workforce to suggest possible defaulters. Dr. Milo, the Chief Data Officer of ME, has decided to replace the manual workforce with a cutting-edge Logistic Regression Model that takes "Age" and "Income" as parameters to predict defaulters. The model is configured such that a "Defaulter" is categorised as the positive class. The final model had the parameters as specified in Table-1 below. To test the model, the data in Table-2 is used as test data. Given this information, answer the given sub questions.

Parameter (Independent variable)	Corresponding coefficient value (Beta value)
Intercept	+3.2
Age (in tens of years)	-1.7
Income (in Lakhs of INR per month)	+1.4

Table-1

Customer ID	Age (in tens of years)	Income (in Lakhs INR per month)	Past Loan Repayment Status
CID01	4.0	0.8	DEFAULTER
CID02	5.3	3.0	DEFAULTER
CID03	2.7	2.5	NOT DEFAULTER
CID04	3.5	2.8	NOT DEFAULTER
CID05	4.7	2.0	NOT DEFAULTER

Table-2

1. [Source Q3; SA]
At a threshold of 0.8, how many "True Positives" is the model predicting?

[Source ans: 1]

2. [Source Q4; SA]
At a threshold of 0.8, how many "True Negatives" is the model predicting?

[Source ans: 0]

3. [Source Q5; SA]
At a threshold of 0.8, how many "False Positives" is the model predicting?

[Source ans: 3]

4. [Source Q6; SA]
At a threshold of 0.8, how many "False Negatives" is the model predicting?

[Source ans: 1]

5. [Source Q7; SA]
At a threshold of 0.5, what is the ACCURACY of the model? (Note: Enter the answer in "%" rounded to two decimal places without the "%" symbol. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 20]

6. [Source Q8; SA]
At a threshold of 0.5, what is the PRECISION of the model for predicting the Negative Class? (Note: Enter the answer in "%" rounded to two decimal places without the "%" symbol. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 0]

7. [Source Q9; SA]
At a threshold of 0.5, what is the RECALL of the model for predicting the Positive Class? (Note: Enter the answer in "%" rounded to two decimal places without the "%" symbol. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 50]

8. [Source Q10; SA]
At a threshold of 0.5, what is the SENSITIVITY of the model? (Note: Enter the answer in "%" rounded to two decimal places without the "%" symbol. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 25]

2024 Sep01 ET

Comprehension for questions 9-15 (source Q29)

A travel company wants to understand its model performance of a classification problem where the task is to classify if its customer will be granted with a tourist visa or not (assume granted = Class 1 and Not Granted as Class 0). Using the table given below, answer the given subquestions

S.No	y_act	y_pred
1	Not Granted	Granted
2	Not Granted	Not Granted
3	Granted	Not Granted
4	Granted	Granted
5	Granted	Granted
6	Not Granted	Granted
7	Granted	Granted
8	Granted	Not Granted
9	Not Granted	Granted
10	Not Granted	Not Granted
11	Not Granted	Not Granted
12	Granted	Granted
13	Granted	Granted
14	Not Granted	Not Granted

9. [Source Q30; SA]
What is the accuracy of the model? Hint: Round your answer to two decimal places and answer them in terms of percentage. Example: If your answers is 0.735, write it as 73.50.

[Source ans: 63.8 to 64.6]

10. [Source Q31; SA]
What is the precision of class 1? Hint: Round your answer to two decimal places and answer them in terms of percentage. Example: If your answers is 0.735, write it as 73.50.

[Source ans: 62.3 to 62.8]

11. [Source Q32; SA]
What is the recall of class 1? Hint: Round your answer to two decimal places and answer them in terms of percentage.Example: If your answers is 0.735, write it as 73.50.

[Source ans: 71.0 to 71.6]

12. [Source Q33; SA]
What is the precision of class 0? Hint: Round your answer to two decimal places and answer them in terms of percentage. Example: If your answers is 0.735, write it as 73.50.

[Source ans: 66.4 to 66.85]

13. [Source Q34; SA]
What is the recall of class 0? Hint: Round your answer to two decimal places and answer them in terms of percentage. Example: If your answers is 0.735, write it as 73.50.

[Source ans: 56.9 to 57.5]

14. [Source Q35; SA]
What is the Sensitivity for the model? Hint: Round your answer to two decimal places and answer them in terms of percentage. Example: If your answers is 0.735, write it as 73.50.

[Source ans: 71.0 to 71.6]

15. [Source Q36; SA]
What is the Specificity for the model? Hint: Round your answer to two decimal places and answer them in terms of percentage.Example: If your answers is 0.735, write it as 73.50.

[Source ans: 56.9 to 57.6]

2024 Dec01 Quiz-2

16. [Source Q3; MCQ]
What is the objective function of logistic regression?

(a) Minimization of squared errors

(b) Maximization of log-likelihood

(c) Minimization of residuals

(d) None of these

[Source ans: b]

17. [Source Q7; MSQ]
What are the applications of logistic regression?

(a) Predicting binary outcomes

(b) Predicting the multi-class output

(c) Predicting the odds of the occurrence of a specific event

(d) None of these

[Source ans: a, b, c]

Comprehension for questions 18-20 (source Q12)

Based on the below confusion matrix, answer the given subquestions. *(Note: Give your answer in decimal (not in %) rounded to two decimal places. For example, if your answer is "0.256", enter it as "0.26")*

		Actual	
		Negative	Positive
Predicted	Negative	145	155
	Positive	75	25

18. [Source Q13; SA]
What is the accuracy of the model?

[Source ans: 0.40 to 0.44]

19. [Source Q14; SA]
What is the precision of the model for predicting the negative class?

[Source ans: 0.47 to 0.49]

20. [Source Q15; SA]
What is the recall of the model for predicting the positive class?

[Source ans: 0.13 to 0.15]

2024 Dec22 ET

Comprehension for questions 21-22 (source Q17)

A logistic model is built to classify three types of animals: "Cat", "Dog" and "Pig". The confusion matrix for the model based on 30 samples is presented in Table-4. Based on the above data, answer the given subquestions

		Predicted		
		Cat	Dog	Pig
Actual	Cat	5	2	3
	Dog	2	5	3
	Pig	4	2	4

Table-4

21. [Source Q18; SA]
What is the accuracy of the model? (Note: Enter your answer in percentage rounded to two decimal places. Eg: If your answer is "10.256%", then enter the answer as "10.26")

[Source ans: 46 to 47]

22. [Source Q19; SA]
What is the precision of the model for predicting "Dog"? (Note: Enter your answer in percentage rounded to two decimal places. Eg: If your answer is "10.256%", then enter the answer as "10.26")

[Source ans: 55 to 56]

Comprehension for questions 23-27 (source Q31)

The table below contains the details of 6 customers who have applied for a bike loan. Based on the 2 inputs (X1 & X2), the bank must decide whether to provide the bike loan or not. In addition, the actual decision on the loan is provided in the column (Y_act). A senior Business Analyst used logistic regression and computed the below values. Based on the data provided answer the subquestions

Cust_id	X1	X2	Y_act
1	3	1.5	1
2	-2	1	1
3	-0.5	-1.5	0
4	1	2.5	0
5	1.65	2.5	0
6	-1.2	0.9	0

β_0	0.3
B_1	0.8
B_2	-0.6
Threshold	0.35

23. [Source Q32; SA]
What is the accuracy of the logistic regression model? Hint: Round off your answer to two decimal places and if your answer is 0.5543, please enter it as 55.43

[Source ans: 33.3 to 33.4]

24. [Source Q33; SA]
What is the precision of class 1? Hint: Round off your answer to two decimal places and if your answer is 0.5543, please enter it as 55.43

[Source ans: 25]

25. [Source Q34; SA]
What is the sensitivity? Hint: Round off your answer to two decimal places and if your answer is 0.5543, please enter it as 55.43

[Source ans: 50]

26. [Source Q35; SA]
What is the precision of class 0? Hint: Round off your answer to two decimal places and if your answer is 0.5543, please enter it as 55.43

[Source ans: 50]

27. [Source Q36; SA]
What is the specificity? Hint: Round off your answer to two decimal places and if your answer is 0.5543, please enter it as 55.43

[Source ans: 25]

2025 Mar16 Quiz-2

Comprehension for questions 28-32 (source Q14)

To automatically detect a card's suit a logistic regression model is developed. For a random sample of 600 cards, the performance of the model is captured by the confusion matrix in Table-1. Given this information, answer the subquestions.

		Predicted Class			
		Hearts	*Spades*	*Diamonds*	*Clubs*
Actual Class	*Hearts*	100	20	40	25
	Spades	15	40	10	35
	Diamonds	20	05	130	05
	Clubs	05	25	15	110

Table-1: Confusion Matrix

28. [Source Q15; SA]
What is the precision for predicting the "Hearts" class? (Note: Enter the answer in percentage rounded to two decimal places. For example, if the answer is "10.235%", then enter it as "10.24")

[Source ans: 71 to 72]

29. [Source Q16; SA]
What is the recall for predicting the "Spades" class? (Note: Enter the answer in percentage rounded to two decimal places. For example, if the answer is "10.235%", then enter it as "10.24")

[Source ans: 39.90 to 40.10]

30. [Source Q17; SA]
What is the sensitivity for predicting "Diamonds" class? (Note: Enter the answer in percentage rounded to two decimal places. For example, if the answer is "10.235%", then enter it as "10.24")

[Source ans: 81.00 to 82.00]

31. [Source Q18; SA]
What is the specificity for predicting "Clubs" class? (Note: Enter the answer in percentage rounded to two decimal places. For example, if the answer is "10.235%", then enter it as "10.24")

[Source ans: 85.00 to 86.00]

32. [Source Q19; SA]
What is the accuracy for the model? (Note: Enter the answer in percentage rounded to two decimal places. For example, if the answer is "10.235%", then enter it as "10.24")

[Source ans: 63.20 to 63.40]

2025 Apr13 ET

Comprehension for questions 33-37 (source Q37)

A fintech company wants to assess the performance of its classification model, which predicts loan approval (positive class) or rejection (negative class). Table 3 presents the data used for model development, while Table 4 provides the estimated coefficients from the logistic regression model. To validate the model, the actual loan decisions made by the fintech company are shown in Table 5. Based on the provided data, answer the given sub-questions

Cust_id	Salary	Age
1	60	30
2	9	35
3	8	40
4	16	25
5	15	50
6	50	21

Table 3: Data used for Building a Logistic Regression Model

Threshold	0.35
b0	0.03
b1	0.04
b2	-0.03

Table 4: Estimated Coefficients from Logistic Regression

Y_act
1
1
0
0
0
0

Table 5: The actual loan decisions

33. [Source Q38; SA]
What is the precision of class 1? (Note: Enter the answer in percentage rounded to two decimal places. For example, if the answer is "10.235%", then enter it as "10.24")

[Source ans: 33.32 to 33.34]

34. [Source Q39; SA]
What is the sensitivity of the model? (Note: Enter the answer in percentage rounded to two decimal places. For example, if the answer is "10.235%", then enter it as "10.24")

[Source ans: 49.95 to 50.05]

35. [Source Q40; SA]
What is the specificity of the model? (Note: Enter the answer in percentage rounded to two decimal places. For example, if the answer is "10.235%", then enter it as "10.24")

[Source ans: 49.95 to 50.05]

36. [Source Q41; MSQ]
What is the correct interpretation of the coefficient b1? (Choose that is/are applicable)

(a) If the salary increases by 1 unit, the log of odds of the application acceptanceincreases by 0.04.

(b) If the salary increases by 1 unit, the odds of the application getting acceptedincrease by 4% (e0.4 = 1.04)

(c) None of these

[Source ans: a, b]

37. [Source Q42; MSQ]
What is the correct interpretation of the coefficient b2? (Choose that is/are applicable)

(a) If the age increases by 1 unit, the log of odds of the application acceptancedecreases by 0.03.

(b) If the age increases by 1 unit, the odds of the application getting acceptеddecrease by 3% (e-0.03 = 0.97).

(c) None of these

[Source ans: a, b]

2025 Aug03 Quiz-2

Comprehension for questions 38-44 (source Q13)

A matrimonial site has built a logistic classification model to predict if a person will get married or not (Married =1 and Not Married =0). The parameters used for this model are Age (in tens of years, that is, the data for a person who is "40 years old" is recorded as "4") and Height (in feet). The built model's coefficients are provided in Table-1. The model is being tested using matrimonial profile data available in Table-2. Given this information, answer the given subquestions

Parameter (Independent variable)	Corresponding coefficient value
Intercept	+8
Age (in tens of years)	-5
Height (in feet)	+2

Table-1

Matrimonial Profile ID	Age (in tens of years)	Height (in feet)	Married or Not Married
CID01	4.0	6	MARRIED
CID02	5.0	6	NOT MARRIED
CID03	2.0	5	MARRIED
CID04	3.5	6.5	MARRIED
CID05	3.0	6.5	NOT MARRIED

Table-2

38. [Source Q14; SA]
At a threshold of 0.8, how many "True Positives" is the model predicting?

[Source ans: 2]

39. [Source Q15; SA]
At a threshold of 0.8, how many "True Negatives" is the model predicting?

[Source ans: 1]

40. [Source Q16; SA]
At a threshold of 0.8, how many "False Positives" is the model predicting?

[Source ans: 1]

41. [Source Q17; SA]
At a threshold of 0.8, how many "False Negatives" is the model predicting?

[Source ans: 1]

42. [Source Q18; SA]
At a threshold of 0.8, what is the ACCURACY of the model? (Note: Enter the answer in "%" rounded to two decimal places without the "%" symbol. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 60]

43. [Source Q19; SA]
At a threshold of 0.8, what is the PRECISION of the model for predicting the Negative Class? (Note: Enter the answer in "%" rounded to two decimal places without the "%" symbol. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 50]

44. [Source Q20; SA]
At a threshold of 0.8, what is the RECALL of the model for predicting the Positive Class? (Note: Enter the answer in "%" rounded to two decimal places without the "%" symbol. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 66.00 to 67.00]

2025 Aug31 ET

Comprehension for questions 45-48 (source Q3)

Given the confusion matrix in Table-1, answer the given subquestions

		Actual Class			
		A	B	C	D
Predicted Class	A	100	0	10	10
	B	10	80	10	25
	C	30	0	70	10
	D	30	10	25	70

Table -1: Confusion Matrix

45. [Source Q4; SA]
What is the number of False Positives for Class-A?

[Source ans: 20]

46. [Source Q5; SA]
What is the precision for predicting Class-D? [Note: Enter your answer in "PERCENTAGE" rounded to two decimal places without the percentage symbol. For example, if your answer is 1.235% then enter the answer as "1.24"]

[Source ans: 50.00 to 52.00]

47. [Source Q6; SA]
What is the recall for predicting Class-B? [Note: Enter your answer in "PERCENTAGE" rounded to two decimal places without the percentage symbol. For example, if your answer is 1.235% then enter the answer as "1.24"]

[Source ans: 87 to 89]

48. [Source Q7; SA]
What is the accuracy of the model? [Note: Enter your answer in "PERCENTAGE" rounded to two decimal places without the percentage symbol. For example, if your answer is 1.235% then enter the answer as "1.24"]

[Source ans: 64 to 66]

Comprehension for questions 49-50 (source Q23)

Assume that you have trained a logistic regression model on a training dataset containing features X1 and X2. The coefficients obtained are given in Table 1. Based on the above data, answer the given subquestions.

β0	0.25
β1	0.3
β2	-0.35

Table 1: Logistic Regression Coefficients

49. [Source Q24; MSQ]
What is the correct interpretation of the coefficient $\beta 1$?

(a) If the X1 increases by 1 unit, the log of odds of the positive outcome increasesby 0.3, assuming X2 to be constant.

(b) If the X1 increases by 1 unit, the odds of the positive outcome increases by35% ($e0.3 = 1.35$), assuming X2 to be constant.

(c) Higher values of X1 are associated with an increased likelihood of the positiveoutcome

(d) None of these

[Source ans: a, b, c]

50. [Source Q25; MSQ]
What is the correct interpretation of the coefficient $\beta 2$?

(a) If the X2 increases by 1 unit, the log of odds of the positive outcome decreases by 0.35, assuming X1 to be constant.

(b) If the X2 increases by 1 unit, the odds of the positive outcome decrease by 29% ($e^{-0.35} \approx 0.71$), assuming X1 to be constant.

(c) Higher values of X2 are associated with a **decreased likelihood** of the positive outcome.

(d) None of these

[Source ans: a, b, c]

51. [Source Q26; MCQ]
Assume that you are working on a classification problem where you are predicting whether the patient is suffering from cancer or not. As a business analyst, select the metric that is most appropriate.

(a) Precision

(b) Recall

(c) R2

(d) MAPE

(e) All of these

[Source ans: b]

52. [Source Q27; MCQ]
Assume that you are working on a classification problem where you are predicting whether the email is spam or not. As a business analyst, select the metric that is most appropriate.

(a) Precision

(b) Recall

(c) R2

(d) MAPE

(e) All of these

[Source ans: a]

2026 Apr06 Quiz-2

Comprehension for questions 53-57 (source Q13)

Dr. Burada is a cardiologist. He is trying to predict if the occurrence of a "Heart Attack" can be predicted using "Age", "Resting bpm" and "Blood Cholesterol Level". Accordingly, the data for 200 patients was collected and a logistic regression model was built (model coefficients are provided in Table-4). As Dr. Burada's wife was running a sweets shop, he wanted to identify patients who were not prone to "Heart Attack" (so that he can sell them more oily snack items and get a name with the Home Minister). Hence, the target is "No Heart Attack=1" and "Heart Attack=0". Table-5 gives the list of patients visiting the sweet shop. Dr.Burada needs to spend approximately 20 mins per Non-Heart Attack patient canvasing if he wants to convince them to buy snacks. Given this information, answer the given subquestions.

Parameter	Value
Intercept	-14
Beta for Age	2.0
Beta for Resting bpm	2.0
Beta for Blood Cholesterol Level	-0.5
Model Threshold	0.8

Table-4: Logistic Model Parameters

Patient-ID	Age	Resting bpm	Cholesterol	Actual Heart Attack
P1	39	80	195	Yes, had an attack
P2	46	95	250	No, did not have an attack
P3	48	75	245	No, did not have an attack
P4	61	65	225	Yes, had an attack
P5	46	85	285	Yes, had an attack

Table-5: List of patients visiting the tea shop

53. [Source Q14; SA]
If Dr. Burada is using the logistic regression model to identify potential people to sell snacks, then what is his accuracy for identifying the potential targets? [Note: Enter your answer in PERCENTAGE rounded to two decimal places without the percentage symbol. For example: If your answer is "1.235%" then enter it as "1.24"]

[Source ans: 39.00 to 41.00]

54. [Source Q15; SA]
If Dr. Burada is using the logistic regression model to identify potential people to sell snacks, what is the sensitivity of the built logistic model? [Note: Enter your answer in PERCENTAGE rounded to two decimal places without the percentage symbol. For example: If your answer is "1.235%" then enter it as "1.24"]

[Source ans: 49.00 to 51.00]

55. [Source Q16; SA]
If Dr. Burada is using the logistic regression model to identify potential people to sell snacks, what is the specificity of the built logistic model? [Note: Enter your answer in PERCENTAGE rounded to two decimal places without the percentage symbol. For example: If your answer is "1.235%" then enter it as "1.24"]

[Source ans: 33.00 to 33.50]

56. [Source Q17; SA]
What is the "odds" for patient "P5" based on the built regression model? [Note: Enter your answer in PERCENTAGE rounded to two decimal places without the percentage symbol. For example: If your answer is "1.235%" then enter it as "1.24"]

[Source ans: 1.40 to 1.80]

57. [Source Q18; MSQ]
If the threshold is decreased to 0.5, which of the following is/are true? (choose all that is applicable)

(a) Precision for predicting Y=1 will increase

(b) Precision for predicting Y=1 will decrease

(c) Recall for predicting Y=1 will increase

(d) Recall for predicting Y=1 will decrease

(e) Precision for predicting Y=0 will increase

(f) Precision for predicting Y=0 will decrease

(g) Recall for predicting Y=0 will increase

(h) Recall for predicting Y=0 will decrease

(i) Accuracy of the model will decrease

(j) Accuracy of the model will increase

(k) None of these

[Source ans: b, f, h, i]

2026 May06 ET

Comprehension for questions 58-64 (source Q12)

The following scenario is purely hypothetical A fintech company wants to assess the performance of its classification model (logistic regression), which predicts loan approval or rejection. Table 2 presents the data used for model development, while Table 3 provides the estimated coefficients from the logistic regression model. To validate the model, the actual loan decisions made by the fintech company are shown in Table 4.

Cust_id	Salary ($1000)	Age
1	55	29
2	11	32
3	9	41
4	17	27
5	18	55
6	55	19

Table 2: Independent Variables

Threshold	0.3
b0	0.04
b1	0.03
b2	-0.05

Table 3: Estimates of Coefficients & Threshold

y_act
1
1
0
0
0
0

Table 4: y actuals

Based on the provided data, answer the given sub-questions.

58. [Source Q13; SA]
What is the accuracy of the model? [Note: Enter your answer in PERCENTAGE rounded to four decimal places without the percentage symbol. For example: If your answer is "81.23445%" then enter it as "81.2345"]

[Source ans: 49.95 to 50.05]

59. [Source Q14; SA]
What is the precision of class 1? [Note: Enter your answer in PERCENTAGE rounded to four decimal places without the percentage symbol. For example: If your answer is "81.23445%" then enter it as "81.2345"]

[Source ans: 33.32 to 33.34]

60. [Source Q15; SA]
What is the sensitivity of the model? [Note: Enter your answer in PERCENTAGE rounded to four decimal places without the percentage symbol. For example: If your answer is "81.23445%" then enter it as "81.2345"]

[Source ans: 49.95 to 50.05]

61. [Source Q16; SA]
What is the precision of class 0? [Note: Enter your answer in PERCENTAGE rounded to four decimal places without the percentage symbol. For example: If your answer is "81.23445%" then enter it as "81.2345"]

[Source ans: 66.6 to 66.7]

62. [Source Q17; SA]
What is the specificity of the model? [Note: Enter your answer in PERCENTAGE rounded to four decimal places without the percentage symbol. For example: If your answer is "81.23445%" then enter it as "81.2345"]

[Source ans: 49.95 to 50.05]

63. [Source Q18; MCQ]
What is the correct interpretation of the coefficient b1?

(a) If the salary increases by 1 unit, the log of odds of the application acceptance increases by 0.03.

(b) If the salary increases by 1 unit, the odds of the application getting accepted increase by 3% (e0.3 = 1.03)

(c) Both If the salary increases by 1 unit, the log of odds of the application acceptance increases by 0.03 and If the salary increases by 1 unit, the odds of the application getting accepted increase by 3% (e0.3 = 1.03)

(d) None of these

[Source ans: c]

64. [Source Q19; MCQ]
What is the correct interpretation of the coefficient b2?

(a) If the age increases by 1 unit, the log of odds of the application acceptance decreases by 0.05.

(b) If the age increases by 1 unit, the odds of the application getting accepted decrease by 5% (e-0.05 = 0.95).

(c) Both If the age increases by 1 unit, the log of odds of the application acceptance decreases by 0.05 and If the age increases by 1 unit, the odds of the application getting accepted decrease by 5% (e-0.05 = 0.95).

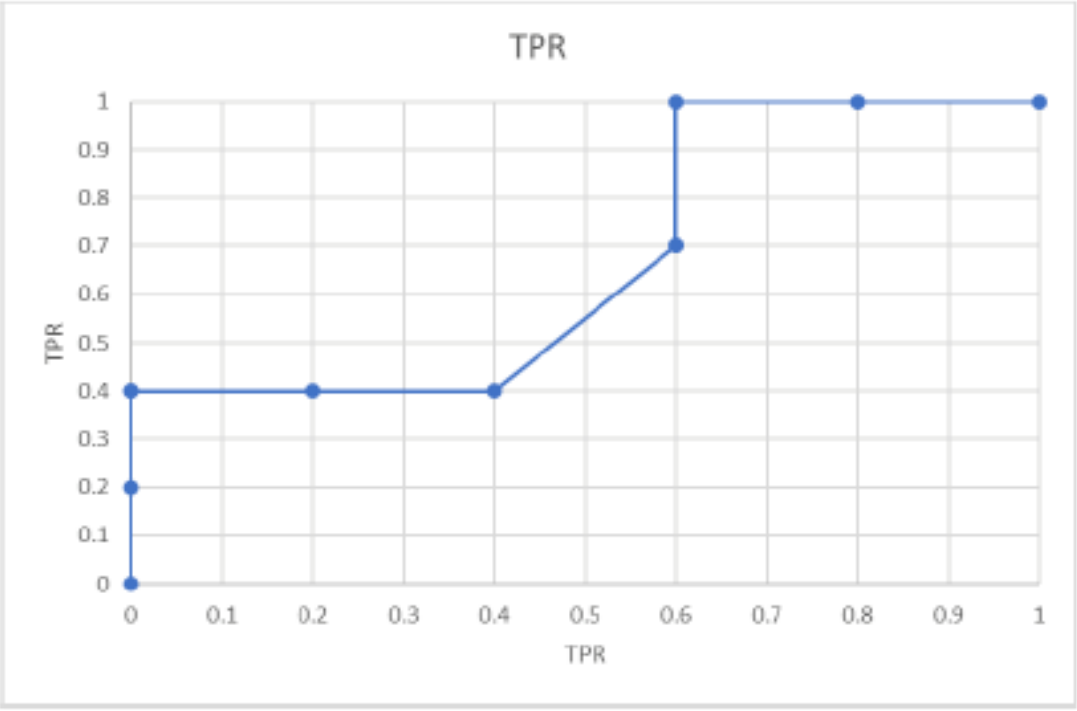
(d) None of these

[Source ans: c]

65. [Source Q20; SA]

Source figure labels both axes TPR. For ROC interpretation, the horizontal axis would normally be FPR; the original figure is preserved.

Calculate the Area under the curve. Hint: Enter your answer in decimals. Round off your answer to 2 decimals. Example: if your answer is 0.7354, write it as 0.74



[Source ans: 0.67]